OSI Model: The Seven Layers of Network Communication

Objective

To understand the seven layers of the OSI (Open Systems Interconnection) model and identify common protocols and technologies associated with each layer.

OSI Model Layer Chart

Layer Number	Layer Name	Main Function	Examples
7	Application	Provides network services to applications	HTTP, HTTPS, DNS, SMTP
6	Presentation	Handles data formatting, encoding, encryption and compression	TLS/SSL, JPEG, UTF-8
5	Session	Establishes, manages and terminates communication sessions	RPC, NetBIOS session services
4	Transport	Provides end-to-end data delivery, segmentation and port-based communication	TCP, UDP
3	Network	Handles logical addressing and routing between networks	IP, ICMP
2	Data link	Transfers frames within a local network using MAC addresses	Ethernet (MAC functions), Wi-Fi (MAC)
1	Physical	Transmits raw bits through a physical	Ethernet physical

		medium or radio signal	standards, fiber-optic cables, radio signals

How Data Travels

When a user opens a website, application data travels down the OSI layers on the sending device and is transmitted over the network. The receiving device processes the data in the reverse order.

Key Protocol Examples

- HTTP/HTTPS: Application layer (Layer 7).
- TCP: Transport layer (Layer 4).
- IP: Network layer (Layer 3).
- Ethernet: Primarily Data Link layer (Layer 2), with Ethernet physical-layer specifications also defined at Layer 1.

Conclusion

The OSI model divides network communication into seven layers. Each layer has a specific responsibility, making network design, troubleshooting and learning easier.

OSI Model – Interview Questions & Answers

1. What is the OSI Model?

Answer: The OSI (Open Systems Interconnection) Model is a conceptual framework that explains how data travels between devices over a network. It consists of seven layers.

2. Which layer uses IP?

Answer: IP (Internet Protocol) works at Layer 3 — Network Layer. It helps route packets from one network to another.

3. Which layer uses TCP?

Answer: TCP (Transmission Control Protocol) works at Layer 4 — Transport Layer. It provides reliable and ordered data delivery.